WRO 2026 *Future Engineers*

Members & coach

Paulina Ibarra Martínez: Member of the Robotics Club at Escuela Normal Superior *"Profr. Moisés Sáenz Garza"* for three years, with participation in two Mexican Robotics Tournaments and a 5th place finish in the most recent edition. Since 2024, serves as a junior coach, achieving third place in WRO 2024 and first place nationally in the Mexican Robotics Tournament 2025.

Christopher Pérez Cortés: Member of the Robotics Club at the *"Prof. Moisés Sáenz Garza"* Teacher Training College since recently, joining the team after completing an intensive robotics course. Currently a third-year middle school student with acquired skills in electronics, programming, and 3D modeling, participating for the first time in a WRO tournament.

Bárbara Daiana García Balboa: Member of the Robotics Club at the Escuela Normal Superior "*Profr. Moisés Sáenz Garza"* for two months, joining the team for a first participation in WRO 2026. This membership follows the completion of two intensive robotics training courses.

Eduardo Alvarado González (coach): Launched in 2014 by an engineering and teaching professional, the *Los Grises Superiores* organization has built a legacy of excellence, securing a series of notable achievements in both domestic and global competitions.

1. *Executive Summary*

This document records the comprehensive development of our autonomous robotic vehicle for the World Robot Olympiad (WRO) competition.

The system is designed to complete three consecutive laps of the circuit autonomously, integrating vision detection algorithms to execute evasion maneuvers: right turns for red obstacles and left turns for green obstacles.

2. Mechanical and Structural Design

Our structure evolved from an initial prototype into an optimized platform for high-level competition.

2.1. Design Transition

We began the project with a model based on LEGO components. Upon evaluating its performance, we determined that the volume and weight were excessive, which limited our maneuverability. Consequently, we replaced the original structure with a custom chassis and component mounts designed by our team and manufactured through 3D printing.

2.2. Advantages of Additive Manufacturing

The implementation of the 3D-printed chassis allowed us to:

- Reduce overall dimensions to facilitate transit through narrow sections.
- Optimize the internal distribution of the Arduino Nano, the PCB, and the power system.
- Improve dynamic stability by precisely adjusting the center of gravity.

2.3. Locomotion and Steering System

The vehicle utilizes four LEGO tires. For trajectory control, we implemented an Ackerman-type steering geometry actuated by a Steren MOT-110 microservomotor, ensuring precise and fluid turns. Traction is provided by a 6V metal gear motor at 97 RPM, achieving an optimal balance between torque and speed.

3. Electronic Architecture and Perception

The central control resides in an Arduino Nano microcontroller, responsible for processing navigation data in real time.

3.1. Proximity Sensors

We installed three HC-SR04 ultrasonic sensors in a vertical orientation. This configuration maximizes precision in distance readings relative to track boundaries and obstacles, eliminating previous detection errors.

3.2. Artificial Vision

For chromatic identification, we integrated an OpenMV camera. Through the development and optimization of image processing algorithms, we achieved fast and accurate color detection, enabling efficient decision-making when identifying obstacles on the track.

3.3. Systems Integration (PCB)

To ensure electrical reliability, we designed and assembled a custom Printed Circuit Board (PCB). This solution eliminated connectivity failures and signal noise present in the initial prototype, consolidating a robust system.

4. Navigation Strategy and Control

We developed software that integrates lane control and vision-based decision-making. The main control system is programmed in C++, responsible for navigation, trajectory adjustment, and overall vehicle behavior, while the vision system is implemented exclusively in Python on the OpenMV camera, where image processing and color detection algorithms are executed. The racing strategy is based on trajectory optimization, allowing the robot to execute laps smoothly by automatically adjusting its speed and steering angle according to each detected obstacle.

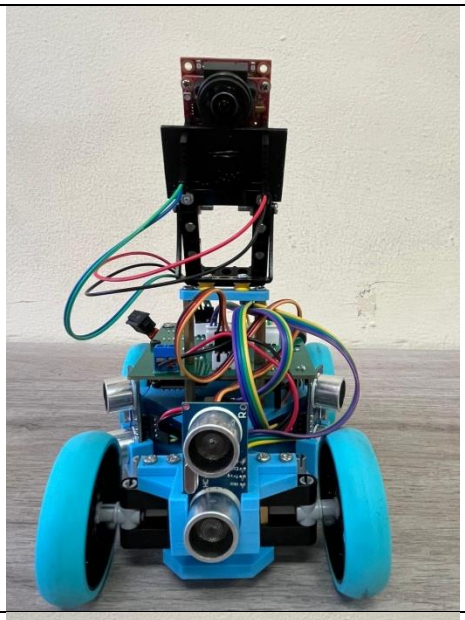
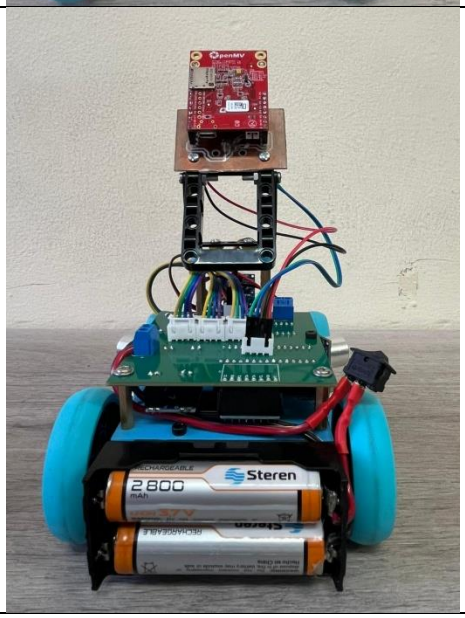
5. Technical Incident Resolution

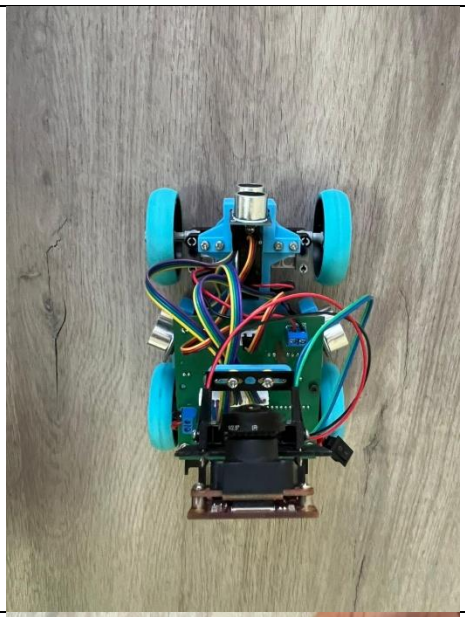
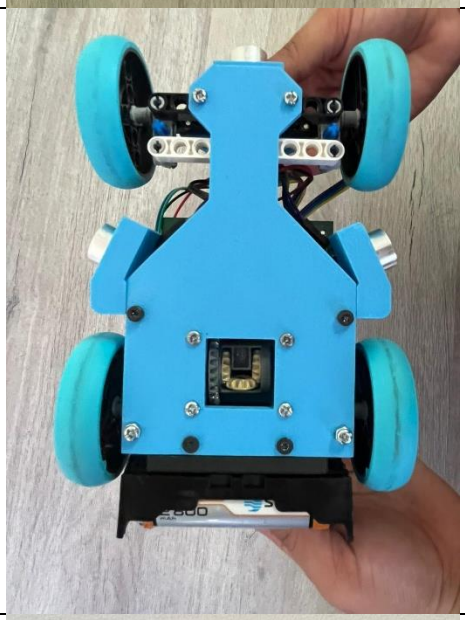
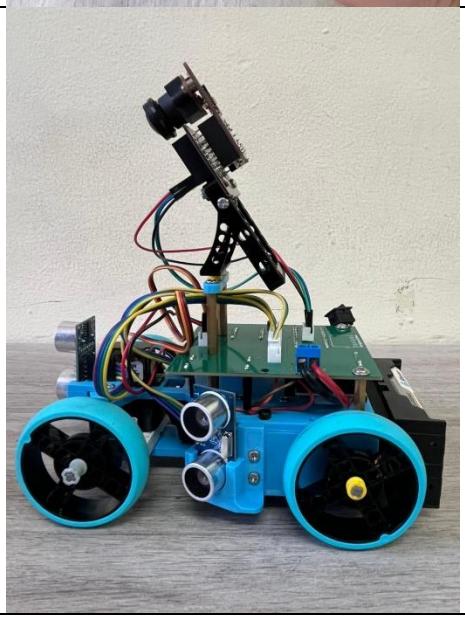
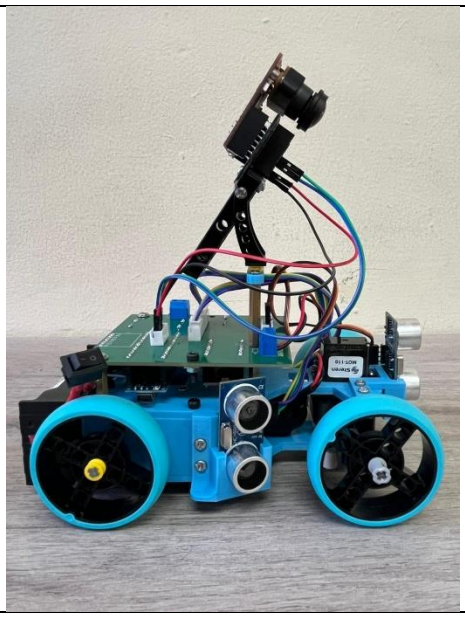
Detected Incident	Technical Impact	Implemented Solution
Excessive dimensions in LEGO chassis	Low efficiency in curves and maneuvers	Replacement with a compact and lightweight 3D chassis.
Signal instability in sensors.	Errors in trajectory maintenance.	Integration of custom PCB and precision soldering.
Limitation in simultaneous color detection.	Erroneous decisions regarding obstacles.	Optimization of firmware and AI processing logic

6.Conclusion

We developed our project from an initial prototype into an efficient and optimized autonomous vehicle. Through continuous improvements in structure, electronics, and programming, we achieved reliable navigation, accurate obstacle detection, and smooth maneuvering. Overall, our work reflects strong problem-solving skills and positions us as a solid competitor for WRO 2026.

7. Robot Photos

Front	
Back	

Top	
Bottom	
Left	

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